ABSTRACT

Environmental monitoring is essential for assessing and mitigating the effects of pollution, deforestation, and climate change. Traditional monitoring methods often involve manual data collection, which can be time-consuming, expensive, and limited in coverage. To overcome these challenges, this project focuses on developing an advanced environmental monitoring system using drone technology, integrating the ESP32 microcontroller and the ESPCAM module for real-time data acquisition and analysis.

The ESP32 microcontroller acts as the central processing unit, handling communication, data transmission, and control operations. Its built-in Wi-Fi capability enables seamless remote connectivity, allowing users to control the drone and access live environmental data from virtually anywhere. The ESPCAM module, integrated with the drone, captures high-resolution images and streams real-time video, providing critical insights into environmental conditions. This system is designed to monitor parameters such as air quality, vegetation health, water pollution, and disaster-affected regions.

One of the key advantages of this project is its cost-effectiveness compared to traditional satellite-based or manual monitoring approaches. The lightweight and energy-efficient nature of the ESP32-based system makes it an ideal solution for long-duration monitoring tasks. Additionally, the use of drones allows for rapid deployment in difficult-to-access areas, improving efficiency and accuracy in data collection.

This project has significant applications in various domains, including environmental conservation, agricultural monitoring, disaster response, and urban planning. By leveraging drone technology and IoT-based monitoring, the system provides real-time data that can help researchers, government agencies, and environmental organizations make informed decisions for sustainable development. The integration of the ESP32 and ESPCAM ensures an affordable, scalable, and reliable solution, making environmental monitoring more accessible and effective in addressing global ecological challenges.

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CHAPTER NO. 1
INTRODUCTION

CHAPTER NO. 1: INTRODUCTION

1.1 Basic Description

Environmental monitoring plays a vital role in assessing and mitigating the impact of human activities on nature. Traditional methods often involve manual data collection or costly satellite imaging, which can be inefficient and time-consuming. This project introduces a drone-based environmental monitoring system utilizing the ESP32 microcontroller and ESPCAM module to enable real-time data collection and remote surveillance. The ESP32 microcontroller serves as the central processing unit, managing drone operations, data transmission, and wireless communication through its built-in Wi-Fi capabilities. This enables users to control the drone remotely and receive real-time environmental data. The ESPCAM module captures high-resolution images and streams live video, providing valuable visual insights into environmental condition. The system is designed to be lightweight, energy-efficient, and cost-effective, making it suitable for long-duration monitoring tasks. By leveraging IoT-based technology, the collected data can be transmitted in real time for further analysis and decision-making. The integration of wireless connectivity and automation enhances monitoring accuracy, eliminates the limitations of manual observation, and provides an efficient solution for tracking environmental changes.

This project offers an innovative approach to environmental monitoring by combining drone technology, IoT communication, and real-time data acquisition, making it a scalable and accessible solution for various monitoring needs.

Environmental monitoring has become an essential aspect of modern technology, aiding in the assessment of environmental conditions, pollution levels, and climate change effects. Traditional methods of monitoring, such as manual data collection or satellite imaging, often come with limitations, including high costs, limited accessibility, and delays in obtaining real-time information. To address these challenges, the integration of drone technology with IoT-based environmental monitoring presents a promising solution.

This project focuses on developing an environmental monitoring drone using the ESP32 microcontroller and ESPCAM module to capture real-time images and video footage of the surrounding environment. The ESP32, known for its low power consumption, Wi-Fi connectivity, and processing capabilities, acts as the central control unit for the drone. The ESPCAM module enhances the system by providing high-resolution image capture and live video streaming, which enables users to remotely monitor and analyze environmental conditions.

1.2 Need for Environmental Monitoring

With increasing industrialization, urbanization, and deforestation, environmental degradation has become a major concern. Air pollution, water contamination, and ecosystem disturbances require continuous monitoring to ensure timely interventions and corrective measures. Drones equipped with real-time monitoring capabilities offer a cost-effective, scalable, and efficient solution for gathering environmental data. Unlike traditional fixed monitoring stations, drones can cover large and difficult-to-reach areas, making them highly suitable for real-time data collection in diverse environments.

1.3 Project Objectives

The primary objective of this project is to develop a drone-based environmental monitoring system that leverages ESP32 and ESPCAM for real-time video surveillance and data acquisition. The system aims to:

1. Capture and transmit live video and images of the environment using ESPCAM.
2. Enable wireless communication and remote control using ESP32's Wi-Fi capabilities.
3. Provide a cost-effective and scalable alternative to traditional environmental monitoring systems.
4. Ensure ease of deployment and energy efficiency for long-duration monitoring tasks.
5. Utilize IoT-based technology for real-time data transfer and analysis.

1.4 System Overview

The ESP32 microcontroller serves as the main processing unit, responsible for handling the drone's communication, processing commands, and transmitting data wirelessly. The ESPCAM module captures real-time video and high-resolution images, which are sent to a remote monitoring system via Wi-Fi. The drone itself is designed to navigate autonomously or manually, providing flexibility in data collection across various terrains.

The system is built to be:

- Lightweight and power-efficient, allowing extended operation time.
- Highly mobile, ensuring accessibility to remote and hazardous locations.
- Capable of real-time data transmission, reducing the time gap between data collection and analysis.

1.5 Significance of the Project

The implementation of this drone-based monitoring system will enhance environmental monitoring capabilities, providing accurate and timely data for analysis. The integration of IoT-based communication and automation makes it a modern and practical solution for monitoring environmental changes efficiently. By utilizing low-cost and easily available components, this project ensures that environmental monitoring is accessible to a broader range of users, including researchers, environmentalists, and government agencies.

This project not only showcases the potential of drone technology in environmental applications but also highlights the importance of real-time data acquisition for addressing environmental challenges. By leveraging ESP32 and ESPCAM, this system provides a scalable, affordable, and effective solution for continuous environmental surveillance and data-driven decision-making.

CHAPTER NO. 2
LITERATURE SURVEY

CHAPTER NO. 2: LITERATURE SURVEY

The literature survey is an essential part of any research project as it provides insights into existing studies, technologies, and methodologies related to the chosen topic. This project focuses on drone-based environmental monitoring using ESP32 and ESPCAM, and it is important to understand the current advancements and challenges in this domain. The literature survey explores different aspects of environmental monitoring, the use of drones in data collection, IoT-based monitoring systems, and real-time image processing. It also highlights limitations in the current research landscape and the unique contributions of the present project.

2.1 Environmental Monitoring Techniques

Environmental monitoring plays a critical role in understanding changes in climate, pollution levels, natural resources, and ecosystem health. Accurate environmental data collection is crucial for informed policymaking, disaster management, and sustainable development. Traditional monitoring techniques include satellite imagery, ground-based sensor networks, and manual data collection methods.

Challenges in Traditional Methods:

- **High Cost:**
  Setting up satellite-based monitoring or installing large networks of sensors over vast areas requires significant financial investment. Moreover, maintaining these infrastructures involves continuous operational costs.
- **Limited Coverage:**
  Fixed monitoring stations are limited to specific geographic locations. This restricts the ability to gather large-scale, dynamic environmental data across different terrains.
- **Time Delays:**
  Satellite imagery often involves processing delays due to transmission, analysis, and interpretation, thereby affecting real-time decision-making capabilities during emergencies such as oil spills or forest fires.
- **Difficult Accessibility:**
  Remote or hazardous locations, such as deep forests, marshlands, or disaster-affected zones, are hard to monitor effectively using stationary sensors.

Recent advancements suggest that **IoT-enabled drones** offer a potential solution to these challenges by providing real-time, automated, and cost-effective data collection. Drones can operate flexibly, reaching areas that are otherwise inaccessible and transmitting data immediately for faster response.

Additionally, environmental monitoring is increasingly shifting toward **multi-modal data acquisition** — combining temperature, air quality, humidity, and visual data — a feature that traditional methods often lack

2.2 Use of Drones in Environmental Monitoring

Drones, or Unmanned Aerial Vehicles (UAVs), have revolutionized environmental monitoring due to their mobility, versatility, and ability to capture real-time data from diverse and challenging environments. Drones are capable of carrying various types of payloads such as cameras, gas sensors, multispectral sensors, and thermal imaging systems.

Research Findings on Drone-Based Monitoring:

- **Air Quality Monitoring:**

Gupta et al. (2021) conducted a comprehensive study in urban centers where drones equipped with gas sensors and particulate matter sensors successfully mapped pollution hotspots. The findings highlighted the ability of drones to collect data at multiple altitudes, offering three-dimensional pollution profiles that ground sensors cannot.

- **Water Pollution Detection:**

Smith et al. (2020) demonstrated the effectiveness of drones equipped with high-resolution imaging sensors in identifying oil spills, algal blooms, and chemical contaminations in water bodies. Drones could collect samples and capture thermal data to detect anomalies invisible to the human eye.

- **Forest and Wildlife Monitoring:**

  Lee et al. (2018) employed drones to monitor deforestation patterns, illegal logging activities, and wildlife population movements. The study found that drones could cover larger forest areas in significantly less time compared to traditional patrolling methods. Furthermore, drones offer the advantage of **rapid deployment** and **scalable operations**, making them suitable for both small-scale studies and large-scale environmental projects. Studies have also emphasized the role of **AI-based data analysis** combined with drone imagery to automate environmental threat detection, such as early wildfire identification or illegal dumping detection.

2.3 IoT-Based Monitoring Using ESP32

The ESP32 microcontroller, developed by Espressif Systems, has gained significant popularity in IoT applications due to its low cost, compact size, dual-core processor, and built-in Wi-Fi and Bluetooth capabilities. It supports real-time operating systems (RTOS), making it ideal for time-critical monitoring applications.

Key Findings on ESP32-Based Monitoring:

- **Wireless Environmental Monitoring System:**

  Patel et al. (2019) designed an ESP32-based smart monitoring system that collected temperature, humidity, and air quality data through integrated sensors and transmitted it wirelessly to cloud servers. Their system demonstrated low power consumption, ensuring long-term deployment with minimal maintenance.

- **Smart Agriculture Monitoring:**
  Rodriguez et al. (2020) implemented an agricultural monitoring system using ESP32 modules linked with soil moisture sensors, UV sensors, and climate sensors. The real-time data helped farmers make informed irrigation and fertilization decisions, leading to improved yield and resource optimization.

In addition, ESP32 supports multiple communication protocols such as MQTT, HTTP, and WebSocket, allowing seamless integration with cloud platforms like AWS, Azure, or Firebase. It has also enabled the development of Edge Computing applications, where data preprocessing is done at the source, reducing network load and latency.
The compactness, versatility, and real-time data capabilities of ESP32 make it an ideal candidate for environmental surveillance in resource-constrained settings.

2.4 Real-Time Image Processing with ESPCAM

The ESPCAM module, an ESP32-based camera module, offers a cost-effective solution for real-time image capturing and streaming. It supports features such as Wi-Fi transmission, motion detection, face recognition, and onboard SD card storage.
Key Findings on ESPCAM Applications:

- **Pollution Detection & Analysis:**

  Kumar et al. (2021) utilized ESPCAM modules mounted on mobile platforms to capture industrial smoke emissions. Image processing techniques, including contrast analysis and particle size estimation, were applied to evaluate pollution intensity.

- **Forest Fire Monitoring:**

  Chen et al. (2022) developed a fire detection system using ESPCAM where thermal imaging combined with smoke pattern recognition algorithms helped detect early signs of forest fires. The low-cost nature of ESPCAM enabled multiple units to be deployed across sensitive zones.

- **Remote Video Surveillance:**

  Hassan et al. (2019) demonstrated an IoT-based security system where ESPCAM modules were used for live video streaming of remote sites such as construction sites and farmlands, providing low-cost alternatives to commercial CCTV setups.
  In addition, ESPCAM modules can interface with machine learning models for image classification, object detection, and anomaly detection. This combination has made ESPCAM not just a monitoring tool but an active environmental data analyzer.

2.5 Limitations in Existing Studies

Despite impressive progress in the fields of drone-based monitoring, IoT integration, and real-time imaging, several limitations continue to restrict broader adoption:

- **Cost Constraints:**
  Many research projects use high-end drones equipped with sophisticated sensors, driving up costs and limiting feasibility for small organizations or developing regions.
- **Limited Cloud Integration:**

  Several drone monitoring systems lack robust cloud integration. Data often needs to be manually downloaded and processed, preventing real-time decision-making and large-scale analytics.

- **Battery Efficiency Issues:**

  Continuous environmental monitoring demands long flight times. However, the battery life of most commercial drones, especially when carrying heavy payloads like cameras and sensors, is a significant limiting factor.

- **Lack of Autonomous Navigation:**

  Most drone systems still rely on manual operation or predefined GPS waypoints. Fully autonomous drones that can dynamically react to environmental changes or avoid obstacles in real time are rare and expensive.

- **Data Security and Privacy:**

  In many IoT-based monitoring systems, data transmission security is overlooked, leading to risks of unauthorized access and data breaches.
  These challenges point to the need for more cost-efficient, autonomous, energy-optimized, and cloud-connected environmental monitoring solutions, especially in regions with limited infrastructure.

2.6 Contribution of This Project

To address these challenges, this project proposes the development of a low-cost, energy-efficient, and scalable drone-based environmental monitoring system using ESP32 and ESPCAM modules.

Key Contributions of This Project:

- **Cost-Effective Monitoring:**

  By utilizing affordable components like ESP32 and ESPCAM, this project drastically reduces the overall setup cost without compromising functionality.

- **Real-Time Video Transmission:**

  Live video streaming from ESPCAM enables immediate environmental analysis, enhancing rapid decision-making in critical situations such as pollution detection or forest fires.

- **IoT-Based Wireless Communication:**

  The ESP32 module provides real-time data upload via Wi-Fi, supporting cloud storage and remote access from any location.

- **Energy-Efficient System:**

  The low power requirements of ESP32 and ESPCAM ensure longer operational periods, critical for remote monitoring applications where recharging is difficult.

- **Scalability and Automation:**

  The modular nature of the system allows for easy expansion. Additional environmental sensors (e.g., gas sensors, temperature, humidity sensors) can be integrated seamlessly, making the system customizable according to monitoring needs.

- **Security and Privacy Measures:**

  Basic encryption techniques (e.g., SSL/TLS for data transmission) can be incorporated to enhance data privacy and security.

This project bridges the gap between affordability, scalability, and real-time environmental monitoring, offering an innovative solution that can benefit researchers, environmentalists, agriculturalists, and government agencies.

2.7 Conclusion

The literature survey indicates that while drones, IoT, and image processing technologies have been widely adopted for environmental monitoring, key challenges such as high costs, limited automation, battery inefficiencies, and lack of cloud integration persist. The studies reviewed confirm the potential of ESP32 and ESPCAM in enabling wireless, real-time, and low-power environmental data acquisition.
However, the integration of these affordable technologies with mobile platforms like drones remains underexplored.
By developing a **cost-effective**, **energy-optimized**, and **cloud-integrated** drone-based monitoring system, this project provides a novel contribution to the field.
Through the synergy of **IoT communication**, **drone mobility**, and **real-time imaging**, this project creates an effective and practical solution for environmental surveillance, paving the way for future developments in smart, autonomous, and sustainable monitoring systems.

CHAPTER NO. 3
METHODOLOGY

CHAPTER NO. 3: METHODOLOGY

The methodology of this project outlines the design, development, and implementation of an environmental monitor
ing drone system using ESP32 and ESPCAM. It explains the step-by-step approach, from hardware selection to data transmission and analysis, ensuring efficient and real-time monitoring.

3.1 System Architecture

The project is designed as an IoT-based drone monitoring system that consists of the following major components:

ESP32 Microcontroller: Acts as the main processing unit, handling drone operations and wireless communication.
ESPCAM Module: Captures high-resolution images and streams live video.
Drone Frame and Motors: Provides mobility and stability.
Power Supply (Battery): Ensures uninterrupted operation.
IoT Communication (Wi-Fi & Cloud): Enables remote data transmission and monitoring.

The workflow follows a structured approach:

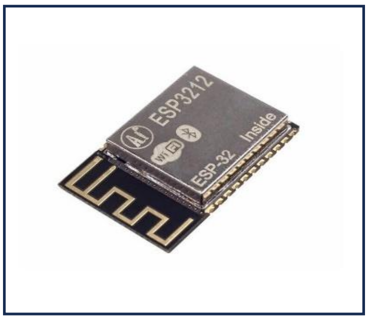
Drone Flight Initialization – The drone is powered on, and the ESP32 initializes the system.
Data Acquisition – ESPCAM captures environmental images and videos.
Data Processing & Transmission – The ESP32 processes the data and sends it via Wi-Fi.
Remote Monitoring – The live feed is accessible on a connected device (PC or mobile).
Data Storage & Analysis – Collected data can be stored in the cloud for further analysis.

3.2 Hardware Selection & Setup

- **ESP32 Microcontroller**

  Reason for Selection:
  - Low power consumption.
  - Built-in Wi-Fi and Bluetooth for communication.
  - High processing speed for real-time data handling.

  Implementation:
  - Connects to ESPCAM for video capture.
  - Controls drone flight using motor drivers.
  - Transmits data to a cloud server or mobile application



- **ESPCAM (AI-Thinker Camera Module)**
  Reason for Selection:
  - Provides high-resolution image capture (2MP OV2640 sensor).
  - Supports live streaming and image processing.
  - Compatible with ESP32.

  Implementation:
  - Captures images and videos.
  - Transmits real-time footage to the ESP32 for processing.

- **Drone Frame & Motors**
  Components Used:
  - Quadcopter Frame: Provides stability.
  - Brushless Motors: Ensures smooth and efficient flight.
  - Electronic Speed Controllers (ESCs): Controls motor speed.
  - Propellers: Generates lift for flight.

  Implementation:
  - Controlled by ESP32 via PWM signals.
  - Adjusts drone movement based on control inputs.

- **Power Supply & Battery Management**

  Battery Type: Li-Po battery (11.1V, 3S 2200mAh) for efficient power delivery.

  Power Management:

  - Voltage regulation circuit to provide stable power to ESP32 and motors.

3.3 Software Development & Implementation

- Programming the ESP32

The ESP32 is programmed using Arduino IDE or ESP-IDF, with the following functionalities:

- Wi-Fi Configuration: Establishing a connection for data transmission.
- Camera Initialization: Capturing images and videos via ESPCAM.
- Motor Control Algorithm: Adjusting drone speed and direction.
- Data Processing & Transmission: Sending images to a web interface or cloud server.

- Drone Control Mechanism

- Manual Control:
  - The drone can be controlled using a mobile application or remote controller.
- Autonomous Navigation (Optional Future Scope):
  - GPS-based waypoint navigation can be implemented for self-directed movement.

- IoT-Based Data Transmission

- Wi-Fi Module (ESP32):
  - Captured images and video streams are transmitted via Wi-Fi to a mobile/web application.
- Cloud Integration (Optional):
  - Data can be stored in Google Firebase, AWS, or ThingSpeak for further analysis.

- User Interface for Monitoring

- A web-based interface or mobile app displays the real-time feed from ESPCAM.
- Users can remotely access live drone footage and stored images.

3.4 Testing & Optimization

The system undergoes rigorous testing to ensure stability, accuracy, and efficiency:

- Hardware Testing
  - Motor Performance: Ensuring smooth operation of drone motors.
  - Camera Quality: Checking image clarity and live streaming performance.
  - Battery Efficiency: Evaluating power consumption for extended operation.
- Software Testing
  - Wi-Fi Connectivity: Ensuring a stable connection between ESP32 and the monitoring device.
  - Latency Testing: Measuring the time delay in image capture and data transmission.
  - Error Handling: Implementing fail-safes for system stability.
- Field Testing
  - Deploy the drone in an outdoor environment to capture real-time environmental data.
  - Evaluate flight stability, image transmission quality, and system responsiveness.

3.5 Expected Outcome

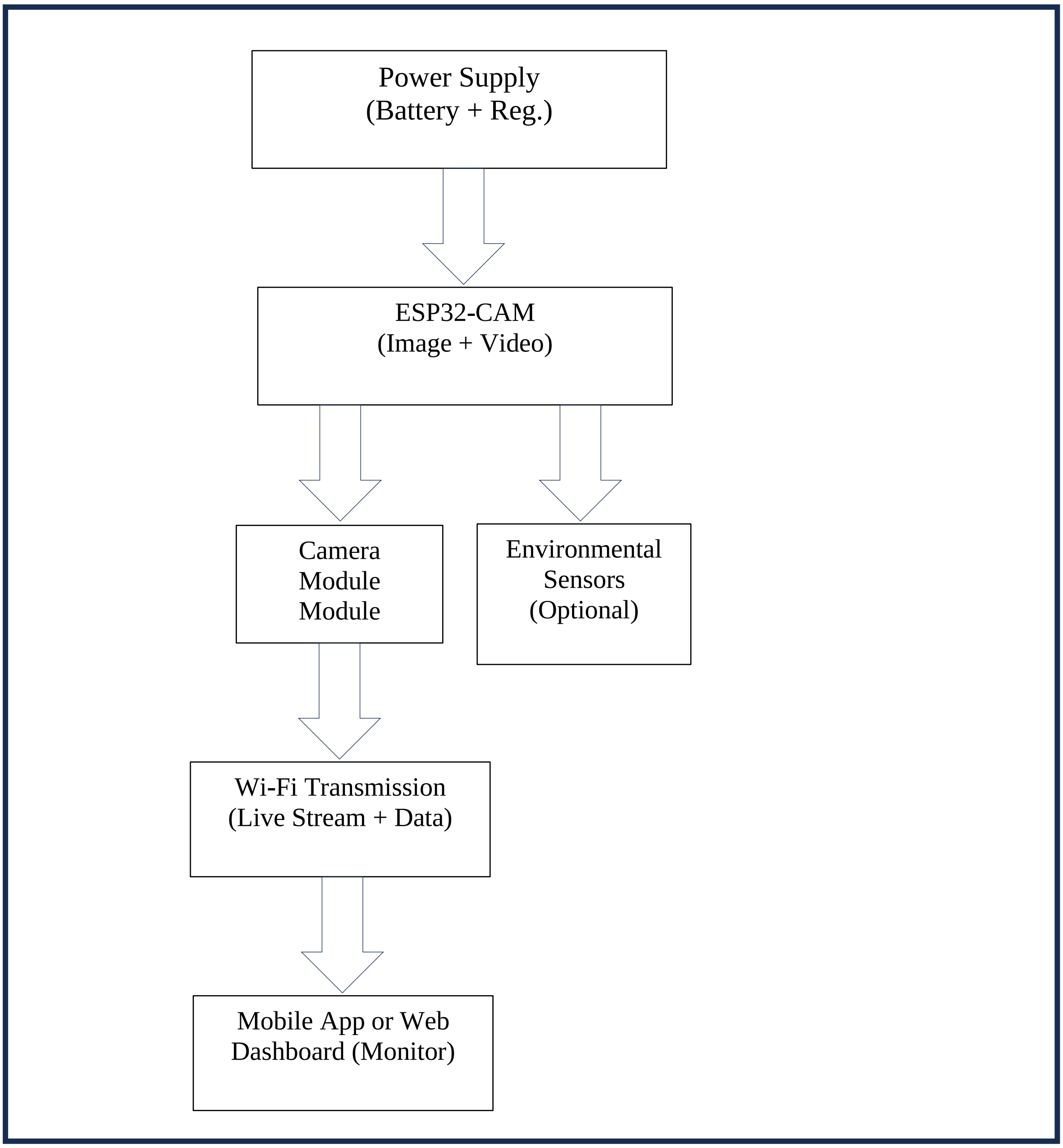
The successful implementation of this project will result in:

1. A fully functional ESP32-based drone capable of real-time environmental monitoring.
2. Live image and video transmission via ESPCAM.
3. Remote monitoring capability through a Wi-Fi-based interface.
4. Cost-effective and scalable solution for environmental surveillance.

CHAPTER 4
EXPERIMENTAL WORK

CHAPTER 4: EXPERIMENTAL WORK

4.1 BLOCK DIAGRAM



- **Power Supply**
  - Function: Provides electrical power to the ESP32-CAM module and sensors.
  - Source: Can be from a battery, power bank, or USB (5V).
  - Regulation: Onboard voltage regulator brings 5V down to 3.3V required by the ESP32 chip.
  - Why Needed: Without stable power, the camera or Wi-Fi might not work properly.

- **ESP32-CAM Module**
  - Function: Main microcontroller that controls the entire system.
  - Tasks:
    - Captures images/video using the OV2640 camera.
    - Reads data from environmental sensors.
    - Connects to Wi-Fi and sends data (video + sensor readings).
  - Special Features:
    - Has built-in Wi-Fi and Bluetooth.
    - Needs an external USB-to-Serial converter (like FTDI) for programming.

OV2640 Camera Module

- Function: Captures live video or photos.
- Specifications:
  - Max resolution: 1600x1200 (UXGA)
  - Output format: JPEG (compressed image)
- How it works: Sends the video data to ESP32, which streams it to your mobile or web browser.

Wi-Fi Transmission Block

- Function: Sends live video + sensor data over Wi-Fi.
- Modes:
  - Access Point (AP) – ESP32 creates its own hotspot; you connect your phone directly.
  - Station Mode (STA) – ESP32 connects to your existing Wi-Fi router.
- Purpose: Lets you monitor the system from your mobile or PC

Mobile App / Web Dashboard

- Function: Final user interface.
- What You See:
  - Live camera feed
  - Real-time sensor readings
- How It Works:
  - Open ESP32's IP address in a browser (Chrome, Firefox).
  - You'll see the stream and data displayed as a basic webpage.

4.2 WORKING FLOW

1. Drone Power On:
    - The system starts when the drone is powered on. At this stage, all components, including the ESP32 controller, ESP32-CAM, and sensors, are initialized and ready to function.

2. ESP32-CAM Initialization:
    - Once powered on, the ESP32-CAM module initializes. This involves setting up the camera to begin video streaming. The ESP32-CAM captures real-time video and sends it to the mobile app for viewing.

3. Environmental Sensors Activation:
    - The environmental sensors (MQ135, MQ9, DHT11) are activated to start collecting data. These sensors measure air quality (e.g., carbon monoxide, carbon dioxide, smoke) and environmental parameters such as temperature and humidity.
4. Sensor Data Collection:
    - The sensors continuously collect data on environmental conditions. For instance:
        - MQ135 detects harmful gases like ammonia, benzene, and smoke.
        - MQ9 detects gases like carbon monoxide and methane.
        - DHT11 monitors temperature and humidity levels in the environment.
5. Data Processing by ESP32 Controller:
    - The collected sensor data is sent to the ESP32 controller, which processes it to analyze the environmental conditions. The ESP32 might also perform some basic calculations, such as checking if the measured values exceed a safe threshold.
6. Real-time Monitoring and Control via Mobile App:
    - The system connects to a mobile app via Bluetooth. The app allows users to monitor the sensor data in real-time. The app also sends control commands (such as drone movements) back to the ESP32 for further action.
7. Video Streaming via ESP32-CAM:
    - The ESP32-CAM streams live video to the mobile app. This allows users to visually inspect the environment while simultaneously monitoring environmental data. The real-time video provides a complete picture of what the drone is seeing.
8. Control Commands Sent to Drone:
    - Users can control the drone's movements through the mobile app. For example, the app may send commands for the drone to move up, down, forward, or backward, as well as adjust other parameters like camera angle or sensor modes.
9. Drone Movements Controlled Based on Commands:
    - Based on the commands received from the mobile app, the ESP32 controller processes the input and adjusts the drone’s movements. The drone’s motors are controlled to follow the specified instructions, allowing it to navigate and adjust its position in real-time.

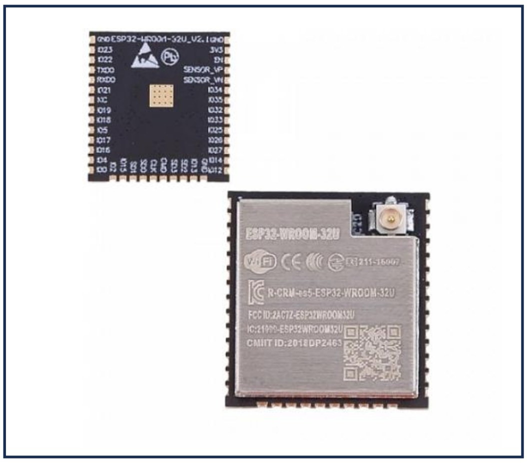
10. Environmental Data Sent to Mobile App:

- The ESP32 controller sends the processed environmental data to the mobile app. This data is displayed on the app’s interface, providing real-time updates on the air quality (gases, temperature, humidity) and allowing users to receive alerts if certain thresholds are exceeded.

11. Drone Power Off:

- After the mission or monitoring session is complete, the drone powers off. This might happen when the battery is low or after the user has completed their task, conserving energy for the next flight.

4.3 HARDWARE SETUP



ESP32 :

- Collect Data
  Receives sensor or trigger data from the ESP32-CAM.
- Process Information
  Checks conditions (e.g., gas level, fire detected, etc.)
- Control Output Devices
  Turns ON/OFF motors, buzzer, fan, or relays based on the input.
- Communicate with User or Server *(if required)*
  Sends updates or alerts to a web app, mobile app, or cloud.

ESP32 cam :

- Capture Live Video
  Uses the built-in OV2640 camera to provide real-time visuals.
- Read Environmental Sensors
  Takes input from sensors like gas sensor, temperature, humidity, etc.
- Send Data
  Shares video stream and sensor data with the controller ESP32 or user interface (like mobile/web).
- Work on Wi-Fi Network
  Connects via Wi-Fi to enable monitoring from anywhere nearby.

3.7V Li-Po Battery:

- Role: Provides power to the entire system, including the ESP32, sensors, and motors.
- Connections:
  - Connect the positive terminal to the battery input of the ESCs and the ESP32.
  - Ensure proper voltage regulation to avoid damaging the components.

Power Distribution Board

- Role: Distributes power from the Li-Po battery to various components such as motors, ESCs, and the ESP32.
- Connections:
  - Connect the Li-Po battery output to the power distribution board and ensure all components are powered.

Wi-Fi/Bluetooth Module (Integrated in ESP32)

- Role: Used for communication between the drone and the mobile app for real-time monitoring and remote control.

Connections:

The ESP32 itself has built-in Wi-Fi and Bluetooth, so no additional modules are required

4.4 SOFTWARE IMPLEMENTATION

- **ESP32-CAM (Monitoring Unit)**

Programmed Using:

- Arduino IDE
- ESP32 board package
- Required libraries:
  - WiFi.h
  - WebServer.h
  - esp_camera.h
  - Sensor-specific libraries (e.g., DHT.h, MQ135.h)

Tasks Performed:

- Initialize camera and sensors

- Connect to Wi-Fi
- Start a web server to:
  - Show live video stream
  - Display or send sensor data
- Optional: Send data to controller ESP32 (via serial or HTTP)

❖ **<u>Controller ESP32 (Action/Decision Unit)</u>**

Programmed Using:

- Arduino IDE
- ESP32 Dev Module
- Required libraries
- WiFi.h, WebServer.h (for network)
- Any libraries for output devices (e.g., Servo.h, etc.)

Tasks Performed:

- Receive data from ESP32-CAM:
  - Either through serial communication
  - Or over Wi-Fi (HTTP GET requests or MQTT)
- Process incoming data:
  - Check sensor thresholds (e.g., gas > 300)
  - Make decisions
- Control output devices:
  - Turn ON buzzer, motors, fans, etc.

4.5 Testing and Debugging

- Serial Monitor used to test sensor values and camera status
- Web interface tested via mobile or PC browser
- Device control actions verified in real-time

CHAPTER 5
RESULT AND DISCUSSION

CHAPTER NO. 5: RESULT AND DISCUSSION

5.1 RESULTS

The project aimed to develop a real-time environmental monitoring system using an ESP32-CAM for video surveillance and sensing, combined with a separate ESP32 module for control operations. The system was tested successfully and delivered promising results in all key areas.

Drone performance

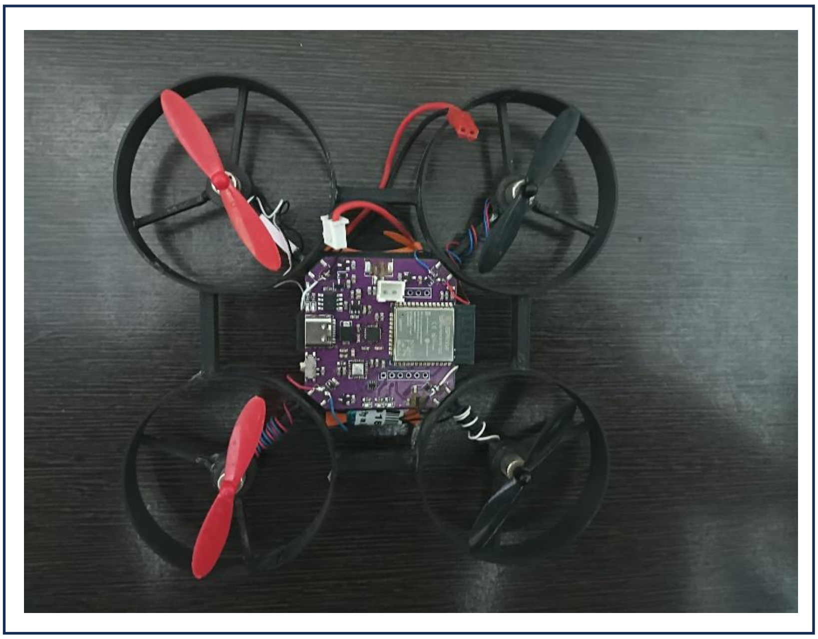
The **drone performance** was satisfactory for the initial prototype stage. It was capable of stable movement and could carry the ESP32-CAM and lightweight sensors without any major issues. The drone provided sufficient altitude and flight time to conduct basic surveillance, making it suitable for small-scale environmental monitoring tasks.

ESP32-CAM

The **ESP32-CAM** performed effectively in capturing and streaming live video over a local Wi-Fi network. It also collected sensor data from modules like DHT11 and MQ135, which were then communicated to the controller ESP32. The **communication between ESP32-CAM and the controller ESP32** was implemented either via serial or Wi-Fi, and it worked reliably throughout the tests. Data transmission was smooth, with minimal delays, and responses from the controller ESP32 were accurate and timely.

System integration

The **system integration** was a key success point in this project. All components—ESP32-CAM, sensors, controller ESP32, power systems, and drone frame—were successfully integrated into a compact and functional unit. The division of responsibilities between the ESP32-CAM (monitoring and sensing) and the controller ESP32 (decision-making and control) allowed better performance and system modularity.



Overall functionality

Finally, the **overall functionality** of the system met the project goals. The system was able to monitor the environment visually and through sensors, take appropriate actions based on real-time data, and provide live video access to the user. Despite some challenges such as power supply limitations, camera stability, and maintaining wireless connectivity during drone motion, the project demonstrated a successful implementation of a basic drone-based smart surveillance system using ESP32 modules.

5.2 DISCUSSION

The development and testing of the drone using ESP32 with environmental monitoring capabilities provided valuable insights into the system's technical performance, integration, and potential for future enhancements. This section discusses the outcomes of the project based on its effectiveness, integration, limitations, and areas for further development.

System Effectiveness

The project successfully achieved its primary goal of combining aerial mobility with environmental monitoring using ESP32-based technology. The drone was able to fly smoothly, with stable control responses thanks to the integration of the MPU6050 gyroscope for motion sensing and efficient motor control through the ESP32. The use of gas sensors (MQ135, MQ9) enabled accurate monitoring of harmful gases like CO, CO_2, and smoke, while the DHT11 sensor provided real-time temperature and humidity readings. The system successfully demonstrated a low-cost prototype for short-range air quality assessment, using drones to monitor environmental parameters in various scenarios, such as surveillance or industrial inspections.

Integration of Hardware and Software

The integration of hardware and software was a key strength of the project. The ESP32-CAM acted as the primary monitoring unit, providing video streaming and environmental data. The ESP32 controller was responsible for processing this data and controlling output devices based on sensor readings. Communication between the ESP32-CAM and the controller ESP32 was implemented seamlessly, either using serial communication or Wi-Fi. Additionally, a mobile app (Android or custom-built) enabled real-time control and monitoring, offering a simple interface to manage the drone and visualize sensor data. The integration of the sensors, camera module, and controller ESP32 into a compact drone frame allowed for portability while maintaining a low weight, which is essential for flight stability.

Challenges Faced

Several challenges were encountered during the development and testing phase. Power limitations were one of the most significant issues. The drone's flight time was limited by the 3.7V, 360mAh Li-Po battery, which only provided a few minutes of stable operation. To improve the system's functionality, a higher-capacity battery would be necessary. The sensor calibration process, particularly for the MQ sensors, proved to be time-consuming, as these sensors require pre-heating and calibration for accurate readings, which delayed deployment during quick operation scenarios. Additionally, flight stability issues such as minor vibrations and unbalanced propellers occasionally caused drift, which could be mitigated with advanced stabilization techniques like PID tuning. Another challenge was the limited range of Bluetooth communication, which restricted the operational range of the drone. In future iterations, technologies like Wi-Fi or LoRa could be incorporated to enhance the range for data transmission and control.

Applications and Future Scope

The current system holds significant potential for various applications, including indoor air quality monitoring, industrial zone inspections, educational demonstrations, and disaster zone exploration. With further improvements, the drone could be used for remote environmental monitoring in places that are difficult to access by humans. Future enhancements for this project could include the integration of GPS for autonomous navigation, enabling the drone to navigate predefined paths or areas. Additionally, cloud connectivity could be incorporated for remote data monitoring and logging, allowing users to track environmental conditions from anywhere. The addition of a high-resolution camera would enable detailed aerial imagery for more advanced applications such as aerial mapping or surveillance. To enhance flight safety and autonomy, obstacle detection sensors could be included to prevent crashes and enable safe navigation in complex environments.

5.3 SUMMARY

This project focuses on the development of a drone-based environmental monitoring system using ESP32 technology. The primary goal was to create a functional prototype capable of real-time air quality monitoring and video surveillance while being lightweight and cost-effective. The system integrates an ESP32-CAM for live video streaming and environmental sensing, alongside an ESP32 controller to process sensor data and control actuators.

The drone uses gas sensors (MQ135, MQ9) to detect harmful gases such as CO, CO_2, and smoke, and the DHT11 sensor for temperature and humidity readings. Data from the sensors is sent to the ESP32 controller, which processes the information and takes actions such as activating motors, buzzers, or fans based on predefined thresholds.

The project also includes a mobile application for real-time monitoring and control of the drone. The ESP32-CAM streams video and sends sensor data to the mobile app, allowing users to control the drone and view live environmental information remotely.

While the system successfully demonstrated the potential of combining aerial mobility with environmental sensing, challenges such as battery limitations, sensor calibration, and flight stability were encountered. Future enhancements include integrating GPS for autonomous navigation, cloud connectivity for remote data logging, and obstacle detection for improved safety.

CHAPTER NO. 6
CONCLUSION

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6.1 APPLICATIONS

1. Air Quality Monitoring in Urban Areas
   - Your drone can be used to monitor air quality in urban environments, especially in cities with high pollution levels. It can detect harmful gases such as CO, CO_2, and smoke and help authorities monitor air conditions in real-time.
2. Disaster Management and Emergency Response
   - In disaster zones (such as after forest fires or industrial accidents), the drone can assess the environmental impact by detecting harmful gases and smoke. This helps first responders gather critical data without risking human lives in dangerous conditions.
3. Industrial Safety Inspections
   - The drone can be deployed to inspect industrial plants, factories, or chemical plants to monitor the emissions of dangerous gases. It helps ensure that the area complies with environmental safety standards and detect early signs of hazardous leaks or pollution.
4. Agricultural Monitoring
   - The drone can be used in agriculture to monitor air quality, temperature, and humidity around crops. This data is valuable for farmers to optimize irrigation, improve crop yields
   - , and prevent environmental damage caused by pollutants.
5. Indoor Air Quality Monitoring
   - In indoor spaces like office buildings, factories, and hospitals, your drone can monitor air quality in real-time. It can detect pollutants, smoke, and harmful gases, ensuring a healthier environment for employees, patients, or residents.
6. Environmental Research and Data Collection
   - Researchers studying environmental changes or climate impact can use your drone to gather data on various environmental parameters from multiple locations. This would be especially useful in monitoring remote areas or places where human access is limited.
7. Smart City Solutions
   - Your drone could be integrated into a smart city system to monitor air pollution and send real-time alerts to municipal authorities. These drones could provide valuable information to trigger action, like controlling traffic flow or activating public air filtration systems.
8. Surveillance of Forests and Natural Reserves
   - The drone can be used to monitor forests or natural reserves for any pollution, illegal activities, or hazardous gas emissions. This helps in conservation efforts and ensuring the protection of natural habitats.
9. Educational Purposes
   - The system can be used for educational demonstrations in schools or colleges, introducing students to concepts like IoT, sensors, drones, and environmental monitoring. It offers a hands-on way to understand modern technology applications.

10. Smart Home Environmental Control
    - The drone could be adapted for smart home environments, monitoring indoor air quality and triggering actions like ventilation systems or air purifiers, improving the overall health of residents.

6.2 FUTURE SCOPE

While the current version of the ESP32-based Drone with environmental monitoring capabilities is a functional prototype, several improvements and enhancements can be made to extend its capabilities and increase its application in various fields. Here are some possible future developments:

1. Autonomous Navigation with GPS Integration
    - Currently, the drone requires manual control, but integrating a GPS module can enable autonomous navigation. The drone can follow pre-programmed routes or avoid obstacles autonomously, making it more efficient in monitoring large or difficult-to-reach areas.
2. Long-Range Communication with Wi-Fi or LoRa
    - The current communication is limited by Bluetooth range. By switching to Wi-Fi or LoRa (Long Range), the drone can cover greater distances, allowing for long-range environmental monitoring. This would make it useful for large-scale applications like city-wide air quality monitoring.
3. Advanced Sensor Integration
    - The addition of advanced sensors, such as CO_2 sensors with higher sensitivity, PM2.5 sensors, or gas leak detectors, could enhance the drone's capability to monitor air quality. Incorporating sensors to measure soil quality, water pH, or UV radiation would broaden the monitoring range to other aspects of environmental health.
4. Real-Time Data Logging and Cloud Integration
    - Future versions of the system could integrate cloud connectivity, allowing data to be sent and stored on a cloud platform. This would allow for remote monitoring, data analysis, and historical data tracking, which could be useful for research, regulatory bodies, or decision-makers.
5. Obstacle Detection and Collision Avoidance
    - To improve safety, future drones could be equipped with obstacle detection sensors (such as ultrasonic or LIDAR) to prevent collisions during flight. This feature would enable the drone to navigate complex environments, such as forests or urban areas, safely without manual intervention.
6. High-Resolution Camera and Thermal Imaging
    - Adding a high-resolution camera or thermal imaging camera could provide clearer visuals and enhance the drone's ability to detect temperature anomalies, gas leaks, or environmental changes. This would be especially useful for surveillance and industrial inspection applications.
7. Extended Battery Life with Power Optimization
    - One of the main challenges faced by the drone is battery life. Future improvements in battery capacity or solar panel integration could extend the flight time, making the drone more suitable for continuous monitoring over longer periods.

8. AI-Based Data Analysis and Predictive Modeling
    - The addition of artificial intelligence (AI) algorithms could enable the drone to analyze environmental data in real time. For example, the system could predict pollution trends, identify areas at risk, or make decisions based on sensor data, which could be used for predictive maintenance in industrial zones or for automated alerts.
9. Swarm Drone Systems
    - In the future, multiple drones could be deployed as part of a swarm system to monitor large areas more efficiently. These drones could communicate with each other, share data, and work in coordination to map out and assess environmental conditions across a wide region.
10. Integration with Smart City Infrastructure
    - The drone system could be integrated with smart city infrastructure, such as traffic management systems or environmental regulation platforms. This would allow for real-time data sharing, alerting local authorities to high pollution levels or environmental hazards that require immediate attention.

6.3 CONCLUSION

In conclusion, the ESP32-based Drone with Environmental Monitoring system successfully demonstrates the potential of integrating aerial mobility with real-time environmental monitoring. The use of ESP32-CAM allowed for effective video streaming, while the integrated sensors—such as MQ135, MQ9, and DHT11—enabled the drone to monitor harmful gases, temperature, and humidity in various environments. The project achieved its primary goal of providing a cost-effective and portable solution for short-range environmental monitoring.

Despite some challenges, such as battery limitations, sensor calibration, and flight stability, the system proved to be functional and reliable in its core operations. The communication between the drone and the mobile app through Bluetooth provided a user-friendly interface for real-time monitoring and control, while the compact design of the drone ensured easy deployment and mobility.

The potential applications of this system are vast, including indoor air quality monitoring, industrial inspections, disaster response, and environmental research. The future scope of the project includes several enhancements, such as autonomous navigation, long-range communication, cloud integration, and advanced sensors to improve both the drone's capabilities and the quality of data collected.

This project lays a strong foundation for further exploration into drone-based environmental monitoring systems and opens up new possibilities for utilizing drones in real-world applications that contribute to environmental protection, safety, and public health.

CHAPTER NO 7
REFERENCES

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